Pilot evaluation: performance matrix

Run `2026-10-11-466b32a`, 111 ProofBench problems, task *prove the given statement*, 2 models.

Table 1: Evaluation setup of run `2026-10-11-466b32a`.

Corpus	`proofbench-statements` at commit `466b32a` (2026-10-08): 111 problems with a main-theorem target; claim status 80 established, 30 open, 1 uncertain. The 28 disproof variants were not evaluated.
Task	Prove the given statement: the model receives the challenge file, whose main theorem ends in a single `sorry`, and must return the completed file with the statement and all definitions unchanged.
Protocol	One request per problem (pass@1), no repair rounds, no tools or retrieval; prompt template `qedrank-prove-v1`.
Models	**Kimi K3**: `kimi-k3` via `api.moonshot.ai`; 2.8 T parameters, 104 B active (mixture of experts); released 2026-07-16; temperature set by the provider. **Qwen3-30B-A3B**: `qwen3-30b-a3b-instruct-2507` via `chat-ai.academiccloud.de`; 30.5 B parameters, 3.3 B active (mixture of experts); released 2025-07-30; temperature 0.
Budgets	16,384 output tokens and 300 s per request.
Grader	Lean v4.34.1, Mathlib `d13f23b`; 400,000 heartbeats, 600 s and 8 GB per compilation. A solution counts as verified when it compiles, contains no `sorry`, keeps the main theorem and every challenge definition unchanged as compiled declarations, uses only `propext`, `Classical.choice` and `Quot.sound`, and replays through the kernel.[a]
Harness	QEDRank commit `65375e7f63`; total wall-clock time 4.3 h.
Statistics	Exact two-sided 95 % Clopper–Pearson intervals. Each problem contributes one item, so clustering by paper does not change the intervals.

[a] This run was graded in the QEDRank default Lean project (`lean`). The benchmark pins Lean v4.33.1, Mathlib v4.33.1, the Matroid library at `fa6c155` and the option `maxSynthPendingDepth = 3`, and its problem library `ProofBench.Mathlib_Problems` was not on the grader's search path.

Table 2: Outcome of every request, by model (111 problems each). Categories are mutually exclusive; each item is assigned to the first step at which it stopped.

Outcome	Kimi K3	Qwen3-30B-A3B
Provider		
Request timed out (300 s)	37 (33.3 %)	0 (0.0 %)
Rate limited (HTTP 429)	57 (51.4 %)	0 (0.0 %)
Grader configuration (not attributable to the model)		
Problem library missing from grader	2 (1.8 %)	48 (43.2 %)
Challenge import rejected by policy	0 (0.0 %)	11 (9.9 %)
Model		
Abstained (no Lean file returned)	15 (13.5 %)	2 (1.8 %)
Lean compile error	0 (0.0 %)	46 (41.4 %) (30 cut off)
Compiled; target statement changed or absent	0 (0.0 %)	1 (0.9 %)
Compiled; proof contains `sorry`	0 (0.0 %)	3 (2.7 %)
Verified	0 (0.0 %)	0 (0.0 %)
Total	111	111

Cut off: the answer reached the 16,384-token output limit before it ended. *Problem library missing*: the challenge imports `ProofBench.Mathlib_Problems`, which the grader's Lean project did not contain. *Challenge import rejected*: the import policy rejected an import that the challenge file itself contains (the Matroid library, `Archive`, or another problem's module); the model added no import.

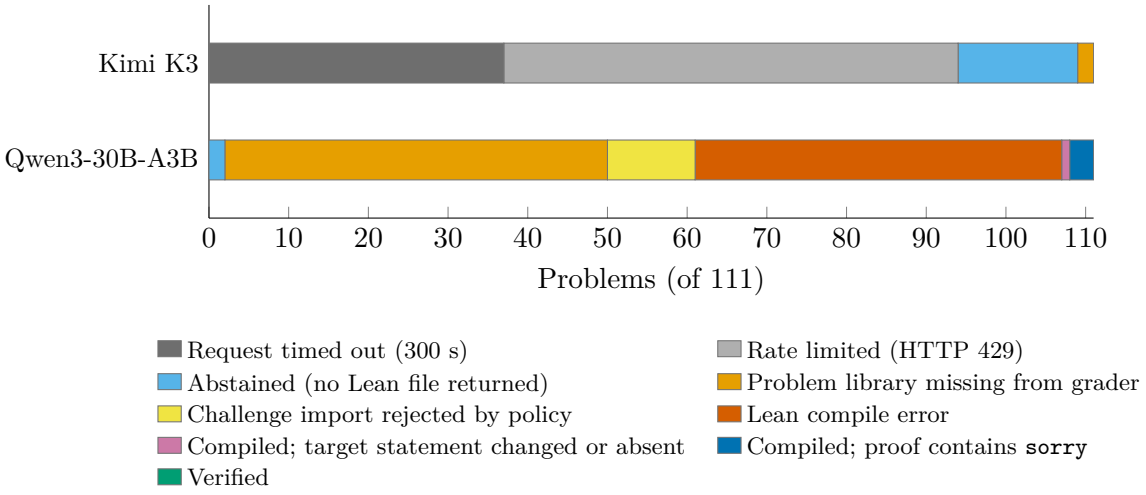


Figure 1: Where each request ended. Grey: provider failures; orange and yellow: answers the grader could not evaluate because of its configuration; the remaining colours are outcomes attributable to the model (Table 2).

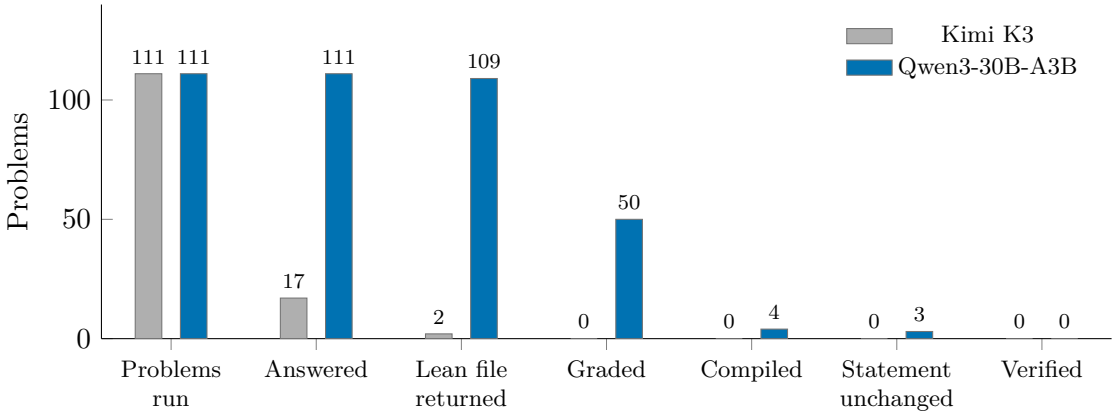


Figure 2: Attrition from request to verified proof. *Graded* counts the returned files that the grader could evaluate; the drop from *Lean file returned* to *Graded* is caused by the grader configuration, not by the model.

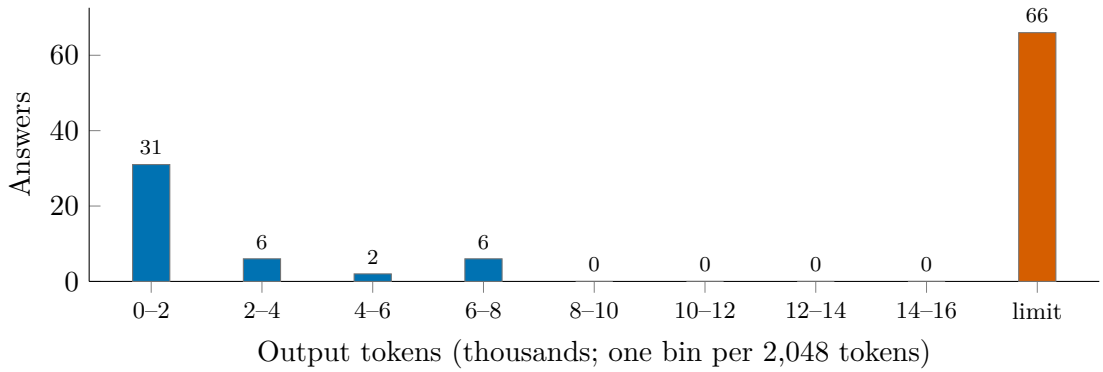


Figure 3: Length of the 111 answers of Qwen3-30B-A3B. 66 answers (59 %, right) reached the 16,384-token limit and were cut off before they ended.

Table 3: Performance matrix on the main theorems (task: prove the given statement, pass@1). Rates exclude provider failures and items the grader could not evaluate.

Measure	Kimi K3	Qwen3-30B-A3B
Problems run	111	111
Requests timed out	37	0
Requests refused (HTTP 429)	57	0
Answered	17	111
Abstained[b]	15/17 (88.2 %)	2/111 (1.8 %)
Answer cut off at token limit	0/17 (0.0 %)	66/111 (59.5 %)
Not gradable (grader configuration)[a]	2	59
Graded proof attempts	0	50
Compile rate[c]	—	4/50 (8.0 %)
95 % CI	—	[2.2, 19.2]
Statement unchanged (of compiled)	—	3/4 (75.0 %)
No `sorry`, standard axioms (of compiled)	—	0/4 (0.0 %)
Verified pass@1[d]	**0/8 (0.0 %)**	**0/43 (0.0 %)**
95 % CI	[0.0, 36.9]	[0.0, 8.2]
Compile rate, as graded[e]	0/2 (0.0 %)	4/109 (3.7 %)
Verified pass@1, as graded[e]	0/8 (0.0 %)	0/80 (0.0 %)
Median latency per answer	146.7 s	81.5 s
Median output tokens per answer	5,444	16,384

[a] The grader could not evaluate the answer: the problem library was missing from its Lean project, or its import policy rejected an import of the challenge file itself (Table 2). These items are excluded from every rate.

[b] Answered without a Lean file. A Lean file that leaves the `sorry` in place is counted as compiled with `sorry`, not as an abstention.

[c] Graded attempts whose file compiled without errors. A compiling file is not a success.

[d] Compiles, contains no `sorry`, keeps the main theorem and all challenge definitions unchanged as compiled declarations, depends only on the standard axioms, and replays through the kernel. Denominator: problems whose claim is established, excluding provider failures and items that could not be graded; abstentions count as failures.

[e] The grader's own figures, before items it could not evaluate are removed; reported for comparison only.
Kimi K3: the 17 answered problems are those that finished within the time limit before the rate limit set in, so its abstention rate is not representative. No comparison between the models is made.

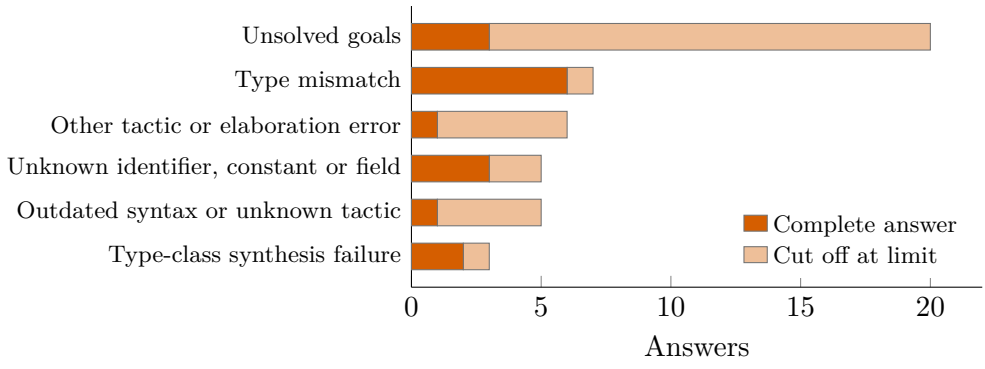


Figure 4: First Lean error of the 46 graded attempts of Qwen3-30B-A3B that failed to compile. Most *unsolved goals* come from answers cut off at the token limit, where the proof stops mid-tactic.

Table 4: Results by the benchmark's difficulty label (Qwen3-30B-A3B; the other model has no graded attempts).

Difficulty label	Problems	Established	Not gradable	Graded	Compiled	Verified
Calibration	12	12	1	11	0	0
Undergraduate	2	2	1	1	0	0
Research	62	62	32	28	1	0
Research, settled Oct 2026	4	4	3	1	0	0
Open	30	0	22	8	3	0
Unlabelled	1	0	0	1	0	0

Calibration merges the benchmark's calibration labels (elementary to upper-end, with or without a Lean reference proof). *Research, settled Oct 2026*: open until October 2026 and settled by a Lean-checked result announced on 6 October 2026 that had not been independently verified when the run was made.

Table 5: Every graded answer that compiled, and why none was verified.

Problem	Model	Claim	Axioms of the main theorem	Outcome
P009	Qwen3-30B-A3B	open	—	main theorem absent
P042	Qwen3-30B-A3B	open	`propext`, `sorryAx`, `Classical.choice`, `Quot.sound`	`sorry` left in proof
P045	Qwen3-30B-A3B	established	`propext`, `sorryAx`, `Classical.choice`, `Quot.sound`	`sorry` left in proof
P077	Qwen3-30B-A3B	open	`propext`, `sorryAx`, `Classical.choice`, `Quot.sound`	`sorry` left in proof

Threats to validity

- **Grader configuration.** Both files returned by Kimi K3 and 59 of the 109 files returned by Qwen3-30B-A3B could not be evaluated, because the grader ran outside the benchmark's pinned Lean project and its import policy rejected imports that belong to the challenge files. The rates in Table 3 therefore rest on 0 graded attempts (Kimi K3) and 50 graded attempts (Qwen3-30B-A3B). The answers that were graded were compiled with Lean v4.34.1 instead of the pinned v4.33.1 and without the benchmark's Lean options.
- **Provider failures.** 94 of 111 requests to Kimi K3 failed (37 timed out after 300 s; 57 were refused with HTTP 429 because the account allows one concurrent request). Its results are not comparable with the other model's.
- **Single attempt.** One request per problem at temperature 0 where the provider allows it, without repair rounds or tools; pass@k and repair success are not measured.
- **Output budget.** 59 % of the answers of Qwen3-30B-A3B reached the 16,384-token limit (Figure 3); a larger budget could change its compile rate.
- **Contamination.** Neither provider states a training cutoff and the problems' publication dates are not recorded, so prior exposure cannot be assessed.